

DOBBIAICO, Italy

WMO: 160330

Lat: 46.733N

Lon: 12.217E

Elev: 1226

StdP: 87.44

Time Zone: 1.00 (EUC)

Period: 94-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.1	-13.0	-21.9	0.6	-3.6	-19.2	0.8	-5.2	8.9	-5.5	7.9	-5.9	0.3	210	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	26.1	14.0	24.8	13.8	23.1	13.3	17.3	21.5	16.3	20.5	15.5	19.7	0.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	13.1	18.7	14.9	12.3	17.9	14.0	11.6	16.9	53.6	21.3	50.5	20.3	47.7	19.7	20.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.2	4.8	3.8	-18.2	28.3	2.6	1.1	-20.1	29.1	-21.6	29.7	-23.1	30.3	-25.0	31.1		
(5)			-18.5	18.6	2.6	1.2	-20.3	19.4	-21.8	20.1	-23.3	20.8	-25.1	21.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.5	-4.1	-2.1	2.3	6.1	11.1	14.6	16.4	15.9	11.4	7.0	1.5	-3.2
	DBStd	8.04	3.90	4.14	4.07	3.45	3.20	3.33	2.69	2.84	3.28	3.66	3.95	4.02
	HDD10.0	1957	437	339	238	122	25	4	0	1	22	104	254	409
	HDD18.3	4356	695	573	497	367	223	116	71	85	208	350	504	668
	CDD10.0	669	0	0	0	6	61	144	198	183	64	13	0	0
	CDD18.3	26	0	0	0	0	1	5	11	9	0	0	0	0
	CDH23.3	273	0	0	0	0	11	68	102	90	2	0	0	0
	CDH26.7	20	0	0	0	0	1	4	7	8	0	0	0	0
(14) Wind	WSAvg	0.7	0.6	0.8	0.9	1.0	0.8	0.7	0.7	0.5	0.6	0.7	0.6	0.6
(15) Precipitation	PrecAvg	957	38	33	50	69	96	113	118	105	87	101	79	48
	PrecMax	1248	121	192	98	162	183	229	195	159	212	302	213	112
	PrecMin	750	0	1	17	30	40	60	36	41	24	10	0	10
	PrecStd	142	26	34	23	33	34	39	37	30	41	69	64	28
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	10.9	15.1	19.7	24.9	27.2	27.8	27.9	23.1	18.5	11.2	6.1
		MCWB	2.8	2.9	5.7	8.7	12.7	13.4	15.3	15.3	13.2	10.8	6.9	4.1
	2%	DB	3.9	7.2	13.1	17.0	22.0	25.7	26.0	25.9	20.9	16.1	9.8	4.2
		MCWB	1.7	1.5	5.0	7.8	11.0	13.1	14.4	14.3	12.3	10.1	6.6	2.7
	5%	DB	2.3	5.2	11.1	15.1	20.0	23.8	24.2	24.0	19.1	14.2	8.1	3.0
		MCWB	0.7	0.8	4.0	6.8	10.3	12.9	14.0	14.2	12.0	9.4	5.7	1.7
	10%	DB	1.2	3.5	9.0	13.1	18.0	21.8	22.9	22.0	17.3	12.7	6.9	1.9
		MCWB	-0.2	0.3	3.4	5.8	9.8	12.3	13.4	13.8	11.5	8.9	5.0	0.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	4.6	7.4	10.7	14.9	17.9	18.8	18.7	16.0	13.4	8.9	4.9
		MCDB	5.0	8.4	12.1	17.3	19.4	22.5	23.5	23.6	19.6	16.3	10.3	5.4
	2%	WB	2.1	2.5	6.1	9.1	13.4	16.3	17.3	17.2	14.8	11.7	7.4	3.2
		MCDB	3.2	5.0	10.4	13.8	18.3	20.5	21.5	21.4	18.2	14.3	8.8	3.7
	5%	WB	1.0	1.7	5.1	8.1	12.3	15.3	16.3	16.2	13.7	10.6	6.3	1.7
		MCDB	2.1	3.9	9.1	12.9	17.0	19.4	20.7	20.2	16.7	13.1	7.6	2.5
	10%	WB	-0.1	0.8	4.2	7.2	11.5	14.2	15.4	15.3	12.7	9.5	5.2	0.7
		MCDB	1.2	3.0	7.7	11.1	15.6	18.2	19.9	19.3	15.7	12.0	6.5	1.5
(35) Mean Daily Temperature Range	MDBR		7.4	9.4	10.1	10.3	10.8	10.8	11.0	10.3	9.9	8.8	7.1	6.2
	5% DB	MCDBR	7.3	11.2	13.4	14.1	14.2	14.3	13.4	13.4	11.9	11.0	8.1	5.6
		MCWBR	5.4	6.4	6.3	6.0	5.5	5.3	4.9	4.9	5.4	5.7	5.3	4.4
	5% WB	MCDBR	6.2	8.8	10.8	11.6	11.7	12.2	11.5	11.1	9.6	8.7	6.3	4.7
		MCWBR	5.0	5.5	5.9	5.5	5.2	5.2	5.0	5.0	5.1	5.1	4.7	4.1
(40) Clear-Sky Solar Irradiance	taub		0.254	0.282	0.329	0.378	0.392	0.403	0.396	0.390	0.356	0.331	0.293	0.257
	taud		2.365	2.300	2.261	2.231	2.259	2.274	2.293	2.329	2.404	2.446	2.487	2.442
	Ebn at Noon		854	892	891	875	873	863	863	855	856	825	802	814
	Edh at Noon		70	93	114	130	131	130	126	117	99	82	64	59
(44) All-Sky Solar Radiation	RadAvg		1.48	2.37	3.50	4.29	4.90	5.52	5.61	4.92	3.70	2.49	1.46	1.24
	RadStd		0.16	0.30	0.42	0.57	0.58	0.46	0.44	0.49	0.42	0.30	0.26	0.17

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	+1.37	+1.33	N/A	N/A	N/A	N/A	
(47) Regional (82 neighbors)		+0.42	N/A	N/A	+0.63	+0.41	+0.40	N/A	-120	+75	+27	

Nomenclature: See separate page